DACUM Live Pro v4.0 - Complete User Guide

The Complete Guide for TVET Professionals

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🎯 Introduction

DACUM Live Pro is a comprehensive web-based tool for creating, verifying, and organizing occupational analysis charts (DACUM charts) with advanced features for live workshop voting and competence clustering.

What You Can Do:

- ✅ Create DACUM charts manually or with AI assistance
- ✅ Conduct live workshop voting with real-time participant engagement
- ✅ Rank tasks by priority using validated formulas

- ✅ Organize tasks into competence clusters for curriculum design
- ✅ Export professional PDF and Word documents

Tool Access: https://hshamjawad.github.io/DACUM-Live-Pro/

🚀 Quick Start: Choose Your Path

Before you begin, decide which workflow fits your needs:

Path A: WITH Live Voting 👥

Best for:

- Facilitated workshops with 8-20 participants
- Group consensus on task priorities
- Real-time data collection
- Validated training needs analysis

Time Required: 4-6 hours (includes workshop)

Path B: WITHOUT Voting 📝

Best for:

- Individual curriculum developer work
- Pre-established task priorities
- Quick chart creation
- Desktop research-based development

Time Required: 2-3 hours

📊 Path Overview: With vs Without Voting

```
┌──────────────────────────────────────────────────────────────┐
│                    START HERE                                 │
└──────────────────────────────────────────────────────────────┘
                          ↓
```

```
┌──────────────────────────────┐
│ COMMON STEPS (Both Paths)    │
│ 1. Chart Info                │
│ 2. Duties & Tasks            │
│ 3. Additional Info           │
└──────────────────────────────┘
          ↓
┌──────────────────────────────┐
↓                              ↓
┌──────────────────┐   ┌──────────────────┐
│ PATH A:          │   │ PATH B:          │
│ WITH VOTING      │   │ NO VOTING        │
├──────────────────┤   ├──────────────────┤
│ • Setup          │   │ • Skip to        │
│ • Participants   │   │   Clusters       │
│ • Vote           │   │   directly       │
│ • Rankings       │   │                  │
└──────────────────┘   └──────────────────┘
│                      │
└──────────────────────────────┘
          ↓
┌──────────────────────────────┐
│ COMPETENCE CLUSTERS          │
│ (Both paths converge here)   │
└──────────────────────────────┘
          ↓
┌──────────────────────────────┐
│ EXPORT (PDF/Word)            │
└──────────────────────────────┘
```

Part 1: Common Steps (Both Paths)

These steps are **required for everyone**, regardless of whether you use voting or not.

Step 1: Chart Info Tab 📋

Purpose: Enter basic information about the occupation and workshop.

What to Fill In:

1. **Occupation Title** ⚠️ *Required*
    - Example: "Industrial Electrician"

- Be specific and use industry-standard terminology

2. **Job Title** ⚠️ *Required*
   - Example: "Entry-Level Electrician"
   - Specify experience level if relevant
3. **DACUM Workshop Date**
   - Enter the date of your workshop or development session
4. **Venue**
   - Location where chart was developed
5. **Produced For**
   - Organization or institution (e.g., "Ministry of Education")
6. **Produced By**
   - Facilitator name(s)
7. **Sector** (Optional but recommended for AI)
   - Example: "Manufacturing", "Healthcare", "Construction"
8. **Context** (Optional but helps AI)
   - Example: "Industrial maintenance in oil & gas sector"
9. **Panel Members** (Optional)
   - List subject matter experts (one per line)
   - Format: Name - Title - Organization
10. **Observers** (Optional)
    - Workshop observers or stakeholders

Logo Upload (Optional):

- **"Produced For" Logo:** Click **"Choose File"** button
- **"Produced By" Logo:** Click **"Choose File"** button
- Accepted formats: PNG, JPG, JPEG
- Logos appear in exported documents

Navigation:

✅ When finished, click the **purple "Proceed to Duties & Tasks"** button at the bottom

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Step 2: Duties & Tasks Tab ✅

Purpose: Build the core DACUM chart by defining duties and tasks.

Key Concepts:

Duty: A major area of work responsibility

- Example: "Install Electrical Systems"

Task: A specific, observable work activity

- Example: "Install conduit and wire in residential buildings"
- Start with **action verbs**: Install, Diagnose, Maintain, Calibrate, etc.

Method 1: AI Generation 🤖 (Recommended for Speed)

Prerequisites:

- ✅ Occupation Title filled in Chart Info
- ✅ Job Title filled in Chart Info
- 💡 Sector and Context help improve AI accuracy

Steps:

1. **Click the green "🤖 Generate AI Draft" button**
    - Located at the top of the Duties & Tasks tab
2. **Wait for generation** (30-60 seconds)
    - Status message appears
    - AI analyzes your occupation and creates duties/tasks
3. **Review Generated Content**
    - AI typically creates 5-8 duties with 4-6 tasks each
    - Check for accuracy and relevance
4. **Edit as Needed:**
    - Click any duty or task to edit text
    - Use **"+ Add Task"** button (green) to add more tasks
    - Use **"+ Add Duty"** button (blue) to add more duties

- Use **red "Remove"** buttons to delete items

✅ **Best Practice:** Always review AI-generated content with subject matter experts

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Method 2: Manual Creation ✏️

1. **Add a Duty:**
   - Click the **blue "+ Add Duty"** button
   - Enter duty name in the text field
   - Example: "Maintain Electrical Equipment"
2. **Add Tasks to Each Duty:**
   - Under each duty, click the **green "+ Add Task"** button
   - Enter task description
   - Example: "Perform preventive maintenance on motor controls"
3. **Repeat** until you have:
   - 5-12 duties (typical range)
   - 3-8 tasks per duty
   - Total: 30-60 tasks (typical for most occupations)

Formatting Tips:

- Use the formatting toolbar to:
  - Add **bullet points**
  - Add **numbering**
  - Apply **bold** or *italic* text
- Keep tasks concise but descriptive

Task Writing Rules:

✅ **DO:**

- Use action verbs
- Be specific and measurable
- Describe observable work
- Example: "Calibrate digital multimeter using known voltage sources"

❌ **DON'T:**

- Use vague language ("understand", "know about")
- Combine multiple tasks
- Example: "Understand electrical principles"

Navigation:

✅ When finished, click the **purple "Proceed to Additional Info"** button

---

Step 3: Additional Info Tab 📚

Purpose: Add supporting information that enriches the DACUM chart.

⚠️ **Note:** All sections are **optional** but highly recommended for comprehensive documentation.

Sections to Complete:

1. Knowledge & Subject Matter

What workers need to **know** (not do):

- Electrical theory
- Safety regulations (OSHA, NEC)
- Blueprint reading
- Mathematics for calculations

Hint: Use bullet points for clarity

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2. Skills Requirements

What workers need to be able to **do**:

- Use hand and power tools
- Read technical diagrams
- Troubleshoot electrical faults
- Work at heights safely

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3. Worker Behaviors & Traits

Personal characteristics for success:

- Attention to detail
- Safety consciousness
- Problem-solving ability
- Teamwork
- Physical dexterity

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4. Tools & Equipment

Physical resources used:

- Multimeters
- Wire strippers
- Conduit benders
- Personal protective equipment (PPE)
- Test equipment

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5. Future Trends & Technology

Industry changes affecting the occupation:

- Smart home automation
- Solar panel integration
- Energy-efficient systems
- Digital controls and IoT devices

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6. Acronyms

Industry-specific abbreviations:

- NEC: National Electrical Code
- GFCI: Ground Fault Circuit Interrupter

- EMT: Electrical Metallic Tubing
- PLC: Programmable Logic Controller

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7. Career Path

Typical progression opportunities:

- Entry Level → Apprentice → Journeyman → Master Electrician
- Specializations: Industrial, Residential, Commercial
- Advancement: Supervisor → Electrical Contractor

Editing Section Headings (Optional):

- Click any section heading to edit it
- Default headings work for most cases

Navigation:

✅ When finished, click the **purple "Proceed to Task Verification"** button

🔀 **DECISION POINT:** You'll now choose between Path A (With Voting) or Path B (Without Voting)

Part 2A: Path WITH Live Voting 👥

Choose this path if you're conducting a facilitated workshop with participants.

Step 4: Task Verification Tab 🎯

Mode Selection:

1. **Data Collection Mode:** Select **"👥 Workshop (Facilitated)"**
    - This enables live voting features
2. **Workflow Mode:** Choose one:
    - **"📊 Standard (DACUM)"** - Most common (uses Importance × Frequency)
    - **"📈 Extended (DACUM)"** - Includes difficulty in calculations

---

Setup Workshop Parameters:

1. Number of Workshop Participants

- Enter total participants (recommended: 8-20)
- This is the maximum responses per task
- Input field appears after selecting Workshop mode

2. Priority Calculation Formula

Choose one:

Option A: Importance × Frequency (I × F)

- Standard DACUM approach
- Best for: Training priority decisions
- Simpler for participants

Option B: Importance × Frequency × Difficulty (I × F × D)

- Extended DACUM approach
- Best for: Identifying training challenges
- More comprehensive analysis

⚠️ **Important:** Choose ONE formula before starting - don't change mid-workshop!

Rating Scales (for Your Reference):

Participants will rate tasks on these scales:

Importance: How critical is this task?

- 0 = Not Important
- 1 = Somewhat Important
- 2 = Important
- 3 = Critical

Frequency: How often is it performed?

- 0 = Rarely
- 1 = Sometimes
- 2 = Often
- 3 = Daily

Difficulty: How hard is it to learn?

- 0 = Easy
- 1 = Moderate
- 2 = Challenging
- 3 = Very Difficult

Step 5: Create Live Voting Session 🔒

Finalize Your Chart:

Click the purple "🔒 Finalize & Create Live Voting Session" button

What Happens:

- ✅ Your duties and tasks are **locked** (no more editing)
- ✅ A unique session ID is created
- ✅ A participant link is generated
- ✅ The Live Workshop section becomes active

⚠️ **Important:** Once finalized, you cannot edit duties/tasks without creating a new session

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Step 6: Share Participant Link 📱

Session Information Displayed:

You'll see a **green box** with:

- ✅ Session ID (for reference)
- 📢 Participant Link (long URL)

Example Participant Link:

https://hshamjawad.github.io/DACUM-Live-Pro/DACUM_LiveWorkshop_Participant.html?lwsession=abc123xyz

How to Share the Link:

Method 1: Copy to Clipboard

- Click the **white "📋 Copy Link to Clipboard"** button
- Paste in email, WhatsApp, Teams, etc.

Method 2: Display on Screen

- Show the link on a projector
- Participants manually type it into their browsers

Method 3: Create QR Code (Recommended)

- Use any free QR code generator online
- Paste the participant link
- Display QR code on screen
- Participants scan with their phones

Step 7: Participants Vote ☑️

⚠️ **This section is for PARTICIPANTS - share this guide with them**

📱 For Participants: How to Vote

Step 1: Open the Participant Link

- Click the link on your device (phone, tablet, or computer)
- **No app or login required**
- Works on any modern browser

Step 2: Review the Occupation

You'll see:

- Occupation Title

- Job Title
- List of Duties with Tasks underneath

Step 3: Rate Tasks

For each task you're familiar with:

1. **Select Importance** (0-3)
    - Click the circle under your chosen rating
2. **Select Frequency** (0-3)
    - Click the circle under your chosen rating
3. **Select Difficulty** (0-3)
    - Click the circle under your chosen rating

💡 **Tips for Participants:**

- ✅ Rate only tasks you **actually know**
- ✅ Think about the **real job**, not ideal scenarios
- ✅ Skip tasks you're unfamiliar with
- ✅ Be **honest** - there are no "right" answers
- ✅ Don't discuss with others while voting

Step 4: Submit Votes

Requirements:

- You must rate **at least ONE task completely** (all 3 dimensions)
- Partial ratings are OK for other tasks

Click the green "✅ Submit All Votes" button

After Submitting:

- Success message appears at the **bottom of your screen** (scroll down)
- Your votes are saved
- **You cannot edit after submission**

⏱️ **Time Required:** 15-20 minutes for most participants

❓ Participant FAQs

Q: Do I need to rate ALL tasks? A: No. Rate only tasks you know. Skip unfamiliar tasks.

Q: Can I change votes after submitting? A: No. Votes are final once submitted.

Q: What if the link doesn't work? A: Contact the facilitator. The session may be closed.

Q: Can I vote on my phone? A: Yes! Fully mobile-friendly.

Q: I refreshed the page, are my votes lost? A: If you already submitted, votes are saved. If voting is still open, you can continue.

Step 8: Monitor Voting Progress 🔄

Back to Facilitator View:

Check Voting Status:

Click the blue "🔄 Refresh Voting Results" button

What You'll See:

- Number of votes received per task
- Preliminary priority rankings
- Mean scores for each dimension
- Progress indicator

💡 **Tip:** Refresh every 5-10 minutes to monitor progress

Step 9: Close Voting Session 🔒

When all participants have finished:

Click the red "🔒 Close Voting (Lock Submissions)" button

What Happens:

- ✅ Voting is **permanently closed**
- ✅ No new votes can be submitted
- ✅ Even if participants refresh the page, they **cannot vote**

- ✅ Results are finalized

⚠️ **Important:** This action cannot be undone. Make sure all participants are finished.

Step 10: Review Final Results 📊

View Rankings:

After closing voting, you'll see:

Priority Rankings Table with columns:

- **Rank** (#1, #2, #3, etc.)
- **Duty** (which duty the task belongs to)
- **Task** (full task description)
- **Mean I** (Average Importance)
- **Mean F** (Average Frequency)
- **Mean D** (Average Difficulty)
- **Priority Index (PI)** - The composite score

Sorting: Tasks are automatically sorted by Priority Index (highest first)

Understanding Priority Index:

If using I × F formula:

- PI = (Mean Importance) × (Mean Frequency)
- Range: 0 to 9
- Higher = More important for training

If using I × F × D formula:

- PI = (Mean Importance) × (Mean Frequency) × (Mean Difficulty)
- Range: 0 to 27
- Higher = More important AND difficult

Step 11: Export Voting Results 📥

You have several export options:

Option 1: Export JSON (Complete Backup)

- Click **purple "📄 Export JSON"** button
- Includes all voting data
- Can be re-imported later
- Preserves everything

Option 2: Export CSV (Spreadsheet)

- Click **green "📊 Export CSV"** button
- Opens in Excel or Google Sheets
- Includes task names and scores
- Good for further analysis

Option 3: Download Pre-Voting Snapshot

- Click **orange "💾 Download Snapshot"** button
- Backup of chart BEFORE voting
- Safety net if you need to start over

Option 4: Export to PDF/Word

- Continue to Part 4: Exporting below

💡 **Best Practice:** Export JSON immediately after voting for backup

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Part 2B: Path WITHOUT Voting 📝

Choose this path if you're NOT conducting live voting.

Skip Task Verification

When you reach the **Task Verification** tab:

Click: "Proceed to Competency Clustering Without Verification"

- This skips the voting setup entirely
- You go directly to Competence Clusters
- Your tasks are not ranked by priority (you can cluster by other criteria)

Continue to: Part 3: Competence Clusters

Part 3: Competence Clusters 🎯

Available for BOTH paths (with or without voting)

Purpose: Group related tasks into logical clusters for curriculum design and training delivery.

What Are Competence Clusters?

Competence Clusters group tasks that share:

- Common skills or knowledge
- Similar workflow or process
- Related work area or equipment
- Logical training sequence

Example:

```
Cluster: "Electrical Safety & Testing"
├─ Inspect personal protective equipment
├─ Test GFCI circuits for proper operation
├─ Verify lockout/tagout procedures
└─ Use voltage testers safely
```

Step 1: Load Ranked Tasks 📊

Navigate to: "🎯 Competence Clusters" tab

Mode Selection (Optional):

Choose your clustering mode:

- 👥 **Session Mode (Collaborative)** - Use during live workshop

- 📝 **Post-Session Mode (Facilitator Only)** - Use for individual refinement

Load Tasks:

Click the purple "📊 Load Ranked Tasks" button

What Happens:

- All tasks appear in a list
- Tasks are sorted by **Priority Index** (if you used voting)
- Each task shows:
  - Task name
  - Original duty
  - Priority indicator (High/Medium/Low)

✅ **Requirement:**

- **Path A users:** Complete voting first
- **Path B users:** Make sure you have duties/tasks created

Step 2: Select Related Tasks ✅

Review the task list and identify tasks that belong together.

How to Select:

1. **Check the box** next to each task you want to include
2. Selected tasks turn **green**
3. You can select tasks from **different duties** - that's perfectly OK!

What to Look For:

Group tasks that share:

- ✅ Similar skills (e.g., all require blueprint reading)
- ✅ Common workflow (e.g., pre-welding preparation steps)
- ✅ Related knowledge (e.g., electrical safety principles)
- ✅ Same work area (e.g., all performed in the shop)

💡 **Example:**

Selected for "Safety & Setup" cluster:
- ☑ Inspect personal protective equipment before use (High Priority)
- ☑ Position workpieces using lifting equipment (Medium Priority)

Step 3: Create the Cluster ✅

Scroll down to: "➕ Create New Cluster" section (green box)

1. **Cluster Name:**
   - Enter a descriptive name
   - Example: "Electrical Systems Maintenance"
   - Be specific and clear
2. **Expert Rationale (Optional but Recommended):**
   - Explain why these tasks belong together
   - Example: "These tasks represent essential safety and preparation steps that must be performed before any welding operation begins."
3. **Click the green "✅ Create Cluster from Selected Tasks" button**

What Happens:

- Cluster is created
- Selected tasks are cleared
- New cluster appears in "Existing Clusters" section below

Step 4: Review Your Clusters 📦

Scroll down to: "📦 Existing Competence Clusters"

Each Cluster Shows:

- **Cluster Name** (in purple/blue heading)
- **Expert Rationale** (if you provided one)
- **Tasks in Cluster:**

- Task name
- Original duty
- Priority ranking (Rank #1, #2, etc.)

Step 5: Edit or Manage Clusters ✏️

For each cluster, you can:

Edit Cluster Name/Rationale:

- Click the **blue "✏️ Edit" button**
- Change name and/or rationale
- Click OK to save

Remove a Task from Cluster:

- Click the **red "✕ Remove" button** next to the task
- Task is removed from cluster
- If it was the last task, you'll be asked to delete the cluster

Delete Entire Cluster:

- Click the **red "🗑️ Delete" button**
- Confirm deletion
- Cluster is permanently removed

Optional: AI-Assisted Suggestions 🤖

Want help creating clusters?

Click the purple "🤖 Suggest Clusters (AI-Assisted)" button (top right)

What Happens:

1. **Confirm the request** (popup appears)
2. **AI analyzes your tasks** (may take 30-60 seconds)
3. **Suggestions appear** in a purple box showing:

- Suggested cluster name
- Rationale
- Number of tasks
- List of tasks included

4. **For each suggestion, you can:**
   - ✅ **Accept** - Creates the cluster automatically
   - ✕ **Close** - Reject the suggestion

⚠️ **Important:** AI suggestions are **advisory only**. You have complete control!

AI Suggestion Example:

```
🤖 AI-Generated Cluster Suggestion 1

Cluster Name: "Pre-Welding Safety & Setup"
Rationale: These tasks represent essential preparation
and safety steps before welding operations begin.

Tasks: 2
- Inspect personal protective equipment before use
- Position workpieces using lifting and positioning equipment

[✅ Accept]
```

💡 **Best Practice:** Review AI suggestions carefully. Accept good ones, modify or reject others.

Best Practices for Clustering 💡

✅ DO:

- Create **3-7 clusters** for most occupations
- Include **3-8 tasks** per cluster
- Focus on **training delivery** - what makes sense to teach together
- Add **rationale** to explain your expert judgment
- Consider **prerequisite relationships** (what should be learned first)

❌ DON'T:

- Create too many tiny clusters (1-2 tasks)

- Create one giant cluster with all tasks
- Ignore tasks from different duties that clearly belong together
- Skip the rationale - it's valuable documentation!

Cluster Naming Tips 📝

Good Cluster Names:

- ✅ "Electrical Safety & Testing Procedures"
- ✅ "Precision Measurement Operations"
- ✅ "Equipment Maintenance & Troubleshooting"

Poor Cluster Names:

- ❌ "Cluster 1"
- ❌ "Miscellaneous Tasks"
- ❌ "Other Stuff"

Good Rationale:

- ✅ "These tasks share common electrical safety knowledge and require similar test equipment skills, making them ideal for a unified training module."

Poor Rationale:

- ❌ "These go together."

Continue to Exporting 📥

Once you've created your clusters, proceed to: **Part 4: Exporting Your Work**

Part 4: Exporting Your Work 📥

Export your complete DACUM chart with voting results and clusters.

Export Options Overview

Located at: Bottom of the page (main export buttons)

Available Formats:

1. 📄 **PDF** - Print-ready, professional format
2. 📝 **Word (DOCX)** - Editable in Microsoft Word
3. 💾 **JSON** - Complete data backup (can reload later)
4. 📊 **CSV** - Spreadsheet (if you used voting)

Export Configuration

Before exporting, choose:

Task Verification Export Mode:

Select ONE radio button:

Option 1: Include Task Verification in DACUM Report (Appendix)

- Exports **ONE file** containing:
  - Complete DACUM chart (duties, tasks, additional info)
  - Task verification results (if you used voting)
  - Competence clusters (if you created them)
  - All in one document with verification as appendix

Best for: Complete documentation in one file

Option 2: Export Task Verification as Standalone Report

- Exports **verification results only**
- No DACUM chart included
- Just the voting/verification data

Best for: Sharing just the priority analysis with stakeholders

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How to Export

Step 1: Select Export Mode (radio button above)

Step 2: Click Export Button

For PDF:

- Click the **blue "Export as PDF"** button
- File downloads automatically
- Filename: [Occupation]_[JobTitle]_DACUM_Chart.pdf

For Word/DOCX:

- Click the **blue "Export as Word (DOCX)"** button
- File downloads automatically
- Filename: [Occupation]_[JobTitle]_DACUM_Chart.docx

What's Included in Exports

Full DACUM Report (Appendix Mode) Includes:

Page 1: Cover/Header

- Occupation Title
- Job Title
- Date, Sector, Context
- Produced for/by with logos

Pages 2-X: DACUM Chart

- All duties in columns
- All tasks under each duty
- Clean, professional layout

Pages X+: Additional Information

- Knowledge Requirements
- Skills Requirements

- Worker Behaviors
- Tools & Equipment
- Future Trends
- Career Path
- Acronyms

Pages Y+: Task Verification Results (If Voting Used)

- Methodology summary
- Number of participants
- Priority formula used
- Complete rankings table:
    - Rank, Duty, Task, Mean I, Mean F, Mean D, Priority Index

Pages Z+: Competence Clusters (If Created)

- Cluster name
- Expert rationale
- Tasks in each cluster
- Original duty for each task
- Priority ranking

All in LEFT-TO-RIGHT format, professional styling

Standalone Verification Report Includes:

Only the verification/voting section:

- Methodology
- Rankings table
- Priority scores

Use when: You only need to share the priority analysis

Additional Export Options

Export JSON (Backup & Reload)

Click: Purple "📄 Export JSON (With Voting Data)" button

What's Saved:

- Complete chart structure
- All duties and tasks
- Voting results (if applicable)
- Clusters (if created)
- All metadata

Use for:

- Backup before making major changes
- Sharing complete data with colleagues
- Re-loading work later

To Reload:

1. Click "📁 Load from JSON" button (top of page)
2. Select your saved .json file
3. Everything restores exactly as saved

Export CSV (Voting Results)

Click: Green "📊 Export CSV (Tasks & Scores)" button

What's Exported:

- Task names
- Priority scores
- Mean ratings
- Spreadsheet format

Use for:

- Further analysis in Excel
- Statistical processing

- Custom reports

File Naming

Automatic filename format:

```
[OccupationTitle]_[JobTitle]_DACUM_Chart.[extension]
```

Example:

```
Industrial_Electrician_Entry_Level_DACUM_Chart.pdf
```

💡 **Tip:** Special characters are replaced with underscores for compatibility

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After Exporting

✅ **Quality Check:**

1. **Open the exported file**
2. **Verify:**
    - All duties and tasks appear correctly
    - Additional information is included
    - Voting results are accurate (if applicable)
    - Clusters are properly formatted (if created)
    - Logos display correctly
3. **If something is wrong:**
    - Return to the tool
    - Make corrections
    - Export again

Sharing Your Work

Who to share with:

- ✅ Curriculum developers
- ✅ Training managers
- ✅ Subject matter experts for review
- ✅ Accreditation bodies
- ✅ Institutional leadership

Recommended format:

- **PDF** for final, read-only distribution
- **Word** when recipients need to edit
- **JSON** when sharing with other DACUM Live Pro users

🔧 Troubleshooting

Common Issues & Solutions

Chart Info Tab

Problem: Can't proceed to next tab
Solution: Make sure Occupation Title and Job Title are filled in

Problem: Logo doesn't appear in export
Solution: Use PNG or JPG format; try re-uploading

Duties & Tasks Tab

Problem: AI generation failed
Solution:

- Check internet connection
- Make sure Occupation Title is filled in
- Wait a moment and try again
- If persistent, create manually

Problem: Can't add more tasks
Solution: Click "+ Add Task" button (green) under the specific duty

Problem: Tasks disappeared
Solution: Check if you accidentally switched tabs; use "Load from JSON" if you have a backup

---

Task Verification / Live Voting

Problem: Can't create voting session
Solution:

- Make sure you have at least one duty with tasks
- Check that you selected Workshop mode
- Verify participant count is entered

Problem: Participant link doesn't work
Solution:

- Copy the FULL link including `?lwsession=`
- Check that voting hasn't been closed
- Try opening in incognito/private mode

Problem: No votes showing up
Solution:

- Click "Refresh Voting Results" button
- Wait for participants to actually submit
- Check that participants used the correct link

Problem: Voting closed but participants still need to vote
Solution: Once closed, you cannot reopen; you'll need to create a new session

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Competence Clusters

Problem: "Load Ranked Tasks" button doesn't work
Solution: Complete voting first (or skip to clusters without voting)

Problem: Can't select tasks
Solution: Click "Load Ranked Tasks" button first

Problem: AI suggestions fail
Solution:

- Check internet connection
- Try manual clustering instead
- Refresh the page and try again

Problem: Clusters don't appear in export
Solution: Make sure you created clusters BEFORE exporting

Exporting

Problem: Export buttons not working
Solution:

- Allow pop-ups in your browser
- Try a different browser (Chrome recommended)
- Disable browser extensions temporarily

Problem: Export is empty or incomplete
Solution:

- Make sure all required fields in Chart Info are filled
- Check that you have duties and tasks created
- Try exporting to a different format

Problem: PDF/Word looks wrong
Solution:

- Update your browser to latest version
- Try clearing browser cache
- Export again

Problem: Can't reload JSON file
Solution:

- Make sure it's a .json file (not .txt)
- File may be corrupted; use backup if available
- Try in a different browser

General Issues

Problem: Tool is slow or freezing
Solution:

- Close other browser tabs
- Clear browser cache
- Refresh the page (WARNING: unsaved work will be lost)
- Save to JSON frequently

Problem: Work disappeared
Solution:

- Check browser history for recent version
- Look for auto-saved JSON exports
- Unfortunately, tool doesn't auto-save - export regularly!

Browser Compatibility

✅ **Fully Supported:**

- Google Chrome (Recommended)
- Microsoft Edge
- Firefox
- Safari (desktop and mobile)

⚠️ **Limited Support:**

- Internet Explorer (not recommended)
- Very old browser versions

💡 **Best Practice:** Use Chrome for best experience

Getting Help

During a Workshop:

- Refer to this guide
- Check the Troubleshooting section
- Ask experienced DACUM facilitators

Technical Support:

Email: hshaam77@gmail.com
LinkedIn: Husham Jawad Kadhim

When reporting issues, include:

- What you were trying to do
- What happened vs. what you expected
- Browser and device you're using
- Screenshots if possible

💡 Tips & Best Practices

Before You Start

✅ **Prepare Subject Matter Experts:**

- Identify 6-12 current workers in the occupation
- Ensure they represent different experience levels
- Brief them on the DACUM process

✅ **Set Up Your Environment:**

- Stable internet connection
- Modern web browser (Chrome recommended)
- Projector for workshops
- Backup device just in case

✅ **Plan Your Time:**

- Chart creation: 3-4 hours
- Live voting: 15-20 minutes
- Results review: 30 minutes
- Clustering: 30-60 minutes

During Chart Creation

✅ **Involve SMEs Throughout:**

- Don't work alone
- Validate every duty and task
- Get consensus on wording

✅ **Save Frequently:**

- Export to JSON every 30 minutes
- Keep multiple backup versions
- Don't rely on browser cache

✅ **Be Specific:**

- Use industry-standard terminology
- Include context where needed
- Avoid vague or generic tasks

During Live Workshops

✅ **Test Everything First:**

- Create a test session
- Try the participant link
- Verify voting works

✅ **Have a Backup Plan:**

- Print paper voting forms

- Have offline version ready
- Prepare for internet issues

✅ **Communicate Clearly:**

- Explain the rating scales
- Show examples
- Answer questions before voting starts

✅ **Monitor Progress:**

- Refresh results every 5-10 minutes
- Help participants who are stuck
- Track completion rate

For Competence Clustering

✅ **Think Training Delivery:**

- What makes sense to teach together?
- What shares common prerequisites?
- What uses similar equipment/facilities?

✅ **Consult Instructional Designers:**

- Get input from curriculum specialists
- Consider learner progression
- Think about assessment strategies

✅ **Document Your Rationale:**

- Explain WHY tasks are grouped
- Note shared competencies
- Record expert consensus

Quality Assurance

✅ **Review with Fresh Eyes:**

- Take a break before final review
- Have someone else read it
- Check for consistency

✅ **Validate with Industry:**

- Share draft with employers
- Get feedback from workers
- Update based on input

✅ **Keep It Current:**

- Review annually
- Update for technology changes
- Revise based on labor market

Documentation Best Practices

✅ **Export Multiple Formats:**

- PDF for distribution
- Word for editing
- JSON for backup

✅ **Version Control:**

- Include date in filename
- Keep previous versions
- Note major changes

✅ **Archive Properly:**

- Store in secure location
- Share with stakeholders
- Make accessible to curriculum developers

📚 Appendix

DACUM Methodology Overview

DACUM = Developing A Curriculum

Core Principles:

1. Expert workers can describe their job more accurately than anyone else
2. All jobs can be effectively described in terms of duties and tasks
3. All tasks have direct implications for curriculum

Traditional DACUM Workshop:

- 1-2 days facilitated session
- 6-12 subject matter experts
- Produces occupational profile
- Forms basis for curriculum design

DACUM Live Pro Advantage:

- Digitizes the entire process
- Adds live voting capability
- Enables priority ranking
- Supports competence clustering
- Professional exports

Glossary

Competence Cluster: Group of related tasks that share common skills, knowledge, or workflow

Duty: Major area of work responsibility within an occupation

DACUM: Developing A Curriculum - a rapid occupational analysis method

Facilitator: Person who guides the DACUM workshop process

I × F Formula: Importance × Frequency (standard priority calculation)

I × F × D Formula: Importance × Frequency × Difficulty (extended priority calculation)

Live Workshop: Real-time participant voting session

Participant Link: Unique URL that allows participants to access voting interface

Priority Index (PI): Composite score calculated from importance, frequency, and difficulty ratings

Session ID: Unique identifier for a live voting session

Subject Matter Expert (SME): Current worker in the occupation being analyzed

Task: Specific, observable work activity

Task Verification: Process of validating and prioritizing tasks through rating

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Workflow Cheat Sheets

Simple DACUM (No Voting)

1. Chart Info → 2. Duties & Tasks → 3. Additional Info →
4. Skip Verification → 5. Competence Clusters (optional) →
6. Export → ✅ Done!

DACUM + Live Workshop

1. Chart Info → 2. Duties & Tasks → 3. Additional Info →
4. Task Verification Setup → 5. Create Session →
6. Share Link → 7. Participants Vote → 8. Close Voting →
9. Review Results → 10. Competence Clusters →
11. Export → ✅ Done!

AI-Powered Fast Track

1. Chart Info (with Sector & Context) →
2. Generate AI Draft → 3. Review & Edit →
4. Additional Info (optional) → 5. Live Workshop →
6. Competence Clusters → 7. Export → ✅ Done!

Quick Reference: Button Colors

🟣 **Purple Buttons:** Navigation / Primary Actions

- "Proceed to Duties & Tasks"
- "Proceed to Additional Info"
- "Finalize & Create Live Voting Session"
- "Load Ranked Tasks"

🟢 **Green Buttons:** Add / Create / Success

- "+ Add Task"
- "+ Add Duty"
- "Create Cluster from Selected Tasks"
- "Submit All Votes" (participant view)

🔵 **Blue Buttons:** Secondary Actions / Refresh

- "+ Add Duty"
- "Refresh Voting Results"
- "Edit" (cluster management)
- "Export as PDF"
- "Export as Word"

🔴 **Red Buttons:** Delete / Remove / Close

- "Remove" (tasks/duties)
- "Delete" (clusters)
- "Remove Task" (from cluster)
- "Close Voting (Lock Submissions)"

🟠 **Orange Buttons:** Backup / Snapshot

- "Download Snapshot (Pre-Voting Backup)"

⚪ **White Buttons:** Copy / Utility

- "Copy Link to Clipboard"

🟣 **Purple AI Buttons:**

- "Generate AI Draft"

- "Suggest Clusters (AI-Assisted)"

Contact & Support

Developer: Husham Jawad Kadhim
Email: hshaam77@gmail.com
LinkedIn: linkedin.com/in/husham-jawad-kadhim-954006200

Tool Access: https://hshamjawad.github.io/DACUM-Live-Pro/

Feedback: Please use the thumbs down button in the interface to report issues or suggest improvements

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Version History

v4.0 (Current)

- Live Workshop voting with real-time participant engagement
- Competence clustering with AI suggestions
- localStorage-based voting lock
- Enhanced exports with verification results and clusters
- Mobile-friendly participant interface

v3.x

- Task verification framework
- Priority ranking formulas
- Enhanced exports

v2.x

- AI generation capability
- Additional info sections

v1.0

- Basic DACUM chart creation
- Manual duty/task entry

🎉 **You're ready to create professional DACUM charts with DACUM Live Pro!**

Remember:

1. Choose your path (with or without voting)
2. Follow the steps in sequence
3. Save frequently (export to JSON)
4. Involve subject matter experts
5. Export when complete

Good luck with your curriculum development! 🚀

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